

一、摘要 Abstract

tw 繁體中文

MCA_WLTNT (Marching Cubes Without Lookup Table and No Triangle) 是一種基於隱式函數幾何重建的新型等值面提取方法。其核心思想為：

不使用 Lookup Table (查表法)，亦不依賴顯式三角形網格建構，而是直接透過隱函數 $f(x, y, z)=0$ 在體素空間中進行幾何點集近似。

此方法將傳統 Marching Cubes 的三個核心步驟：

拓模分類 (Lookup Table)

邊界交點計算

三角網格生成

重新改寫為：

符號變化檢測 (Sign-based detection)

邊緣交點直接插值 (Edge interpolation)

點雲式曲面近似 (Point-based surface representation)

最終結果不再強制生成 triangle mesh，而是：

以高密度幾何點集近似連續曲面。

GB English Abstract

MCA_WLTNT (Marching Cubes Without Lookup Table and No Triangle) is a geometric implicit surface reconstruction framework based on scalar fields.

Instead of relying on:

Lookup tables (topological cases)

Explicit triangle mesh construction

it directly operates on the implicit function:

$$f(x, y, z)=0$$

within voxelized space.

The traditional pipeline:

Case classification (lookup table)

Edge intersection computation

Triangle mesh generation

is replaced by:

Sign-based field evaluation

Direct edge interpolation

Point-based surface approximation

The final output is no longer a triangle mesh, but:

a dense geometric point set approximating a continuous implicit surface.



二、核心數學基礎 Mathematical

Foundation

TW

MCA_WLTNT 建立於隱式曲面：

$$f(x, y, z)=0$$

在每個 voxel（單位立方體）中，我們檢查八個頂點：

若 $f(P_i) \cdot f(P_j) < 0$ 則稜線(點 i & 點 j)邊上存在交點

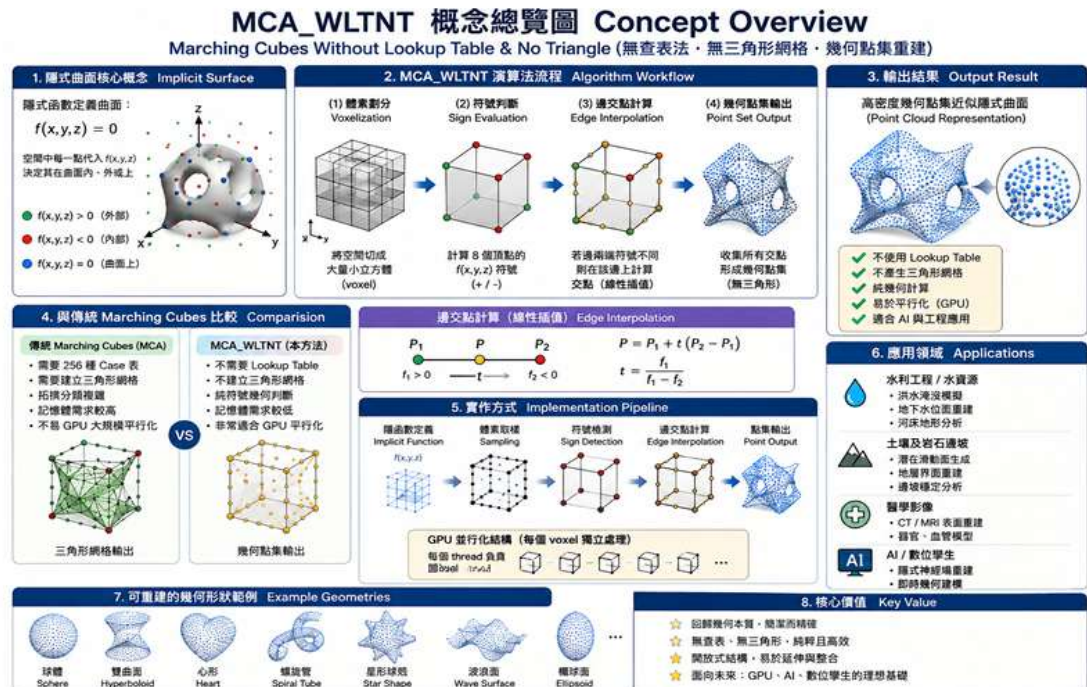
交點以線性插值計算：

$$P = P_1 + t(P_2 - P_1)$$

$$t = f_1 / (f_1 - f_2)$$

但與傳統 MCA 不同：

✗ 不進行三角形重建



✓ 直接輸出交點集合

GB

The method is based on the implicit surface:

$$f(x, y, z) = 0$$

For each voxel, the algorithm evaluates 8 corner values.

If:

$$f(P_i) \cdot f(P_j) < 0$$

an intersection exists along the edge.

The intersection point is computed by linear interpolation:

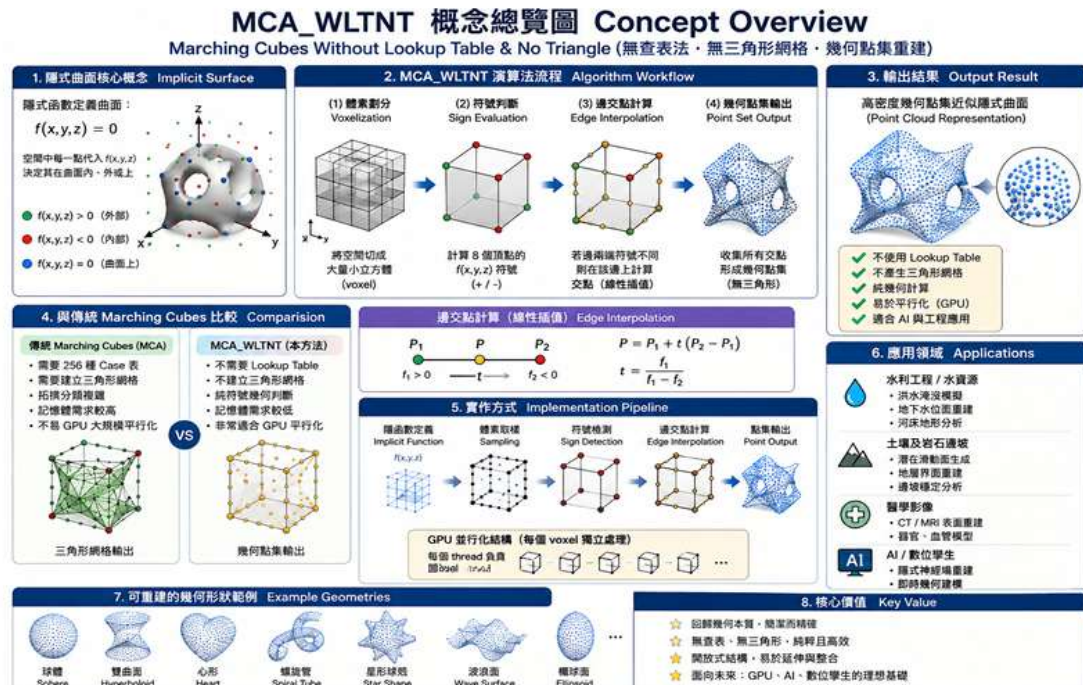
$$P = P_1 + t(P_2 - P_1)$$

$$t=f1/(f1-f2)$$

However, unlike classical Marching Cubes:

✗ No triangle reconstruction

- ✓ Direct point set generation



📌 三、MCA_WLTNT 核心思想 Core Concept

TV

MCA_WLTNT 的核心突破在於：

🧠 1. 去 Lookup Table 化

不再使用 256 case table，而是直接由幾何判斷生成結果。

2. 去三角網格化

不強制 polygon mesh，而是：

以 point cloud 表示等值曲面

3. 幾何極限思想

當 voxel 足夠小：

$dx, dy, dz \rightarrow 0 \implies dx, dy, dz \rightarrow 0$

點集合極限趨近真實曲面。

GB

The key innovations are:

1. No Lookup Table

No predefined topology classification is required.

2. No Triangle Mesh

Surface is represented as a point cloud instead of polygons.

3. Geometric Limit Interpretation

As voxel size approaches zero:

$dx, dy, dz \rightarrow 0 \implies dx, dy, dz \rightarrow 0$

the point set converges to the true implicit surface.



四、與傳統 Marching Cubes 差異

Comparison

項目	Traditional MCA	MCA_WLTNT
Lookup Table	必須	不需要
Triangle Mesh	必須	不需要
Representation	Mesh	Point cloud
Complexity	高	較低

項目	Traditional MCA	MCA_WLTNT
GPU Friendly	中等	高
AI integration	困難	容易

五、數值與幾何特性 Properties

TW

- ✓ 自然幾何生成
 - ✓ 無拓撲分類問題
 - ✓ 易 GPU parallelization
 - ✓ 適合 AI implicit learning
 - ✓ 適合 CT / MRI / LiDAR
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GB

- ✓ Pure geometric formulation
 - ✓ No topology ambiguity
 - ✓ GPU-friendly parallel structure
 - ✓ Compatible with AI implicit models
 - ✓ Suitable for CT / MRI / LiDAR reconstruction
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六、從 Marching Squares 到 MCA_WLTNT

TW

2D 情況：

Marching Squares → contour line

3D 擴展：

MCA → surface mesh

MCA_WLTNT → point-based surface

GB

2D case:

Marching Squares → contour lines

3D extension:

MCA → triangle mesh surface

MCA_WLTNT → point-based implicit surface

七、工程應用 Engineering Applications

TW

 水利 / 水資源

洪水淹沒模擬

地下水位面

河床地形重建

 邊坡工程

滑動面自動生成

地層界面分析

安全係數輔助

 醫學影像

CT / MRI surface reconstruction

血管建模

 AI / Digital Twin

implicit neural field

real-time geometry modeling

GB

 Hydrology

Flood simulation

Groundwater surface modeling

Riverbed reconstruction

 Geotechnical engineering

Automatic slip surface detection

Stratigraphic modeling

Stability analysis support

 Medical imaging

CT / MRI reconstruction

Vascular modeling

 AI / Digital Twin

Neural implicit fields

Real-time geometry systems



八、Python / GPU 延伸 Python & GPU

Extension

TW

MCA_WLTNT 適合：

NumPy vectorization

CuPy GPU acceleration

CUDA compute kernels

PyTorch geometric learning

GB

MCA_WLTNT is suitable for:

NumPy-based implementation

GPU acceleration (CuPy / CUDA)

Parallel voxel processing

AI-based geometry learning



九、結論 Conclusion

TW

MCA_WLTNT 提供一種新的幾何視角：

不再依賴查表與多邊形網格，而是回到「隱式函數本身的幾何極限」。

它將 Marching Cubes 從：

engineering lookup system

提升為

geometric implicit reconstruction system

GB

MCA_WLTNT introduces a new geometric paradigm:

abandoning lookup tables and triangle meshes, returning to the pure geometry of implicit functions.

It transforms Marching Cubes from:

an engineering lookup-based system

into

a geometric implicit reconstruction framework.

十、引用 Citation

C. H. Day.

MCA_WLTNT: Geometric Marching Cubes Without Lookup Tables and No Triangle Representation.

GitHub Research Note, 2026.